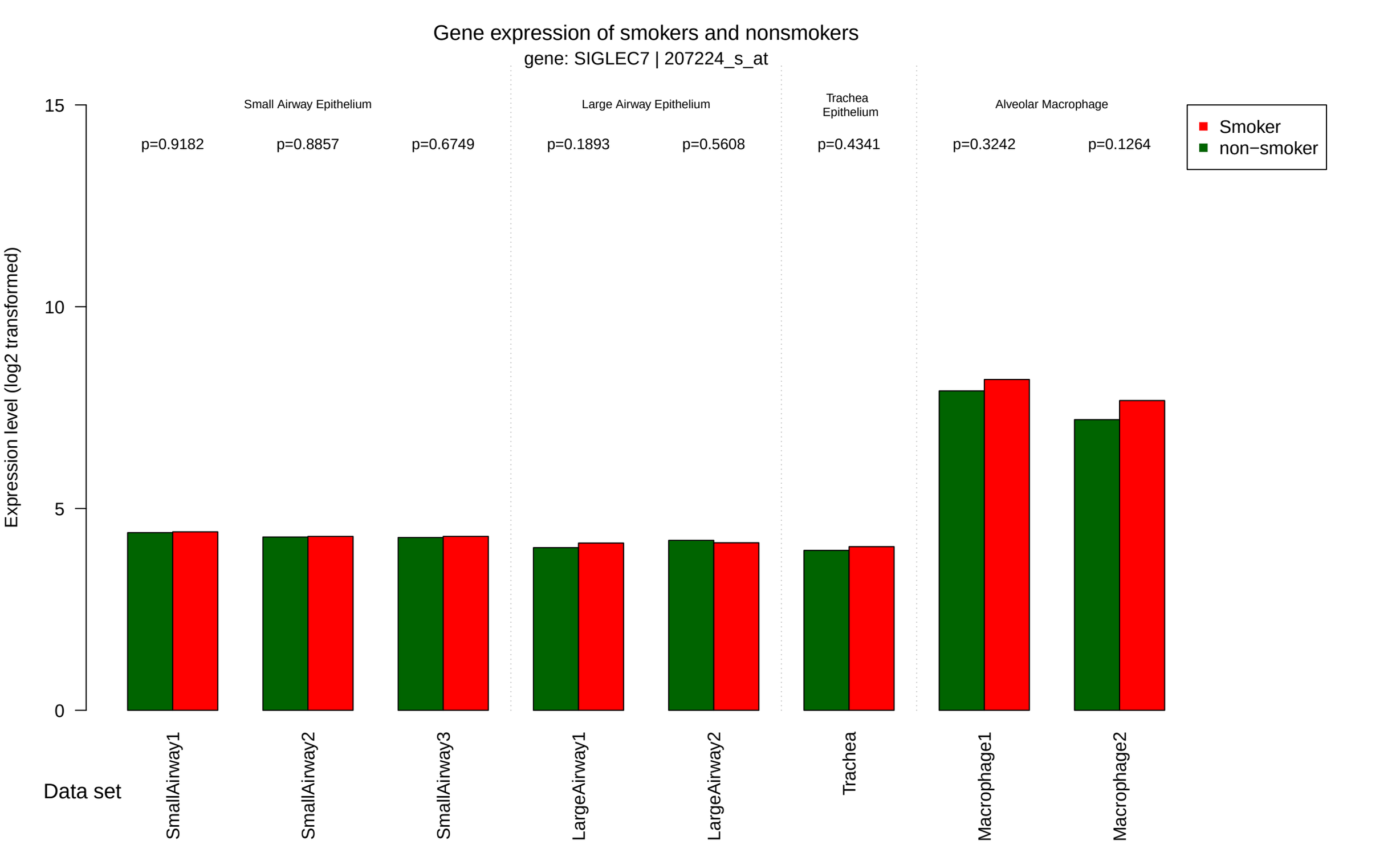
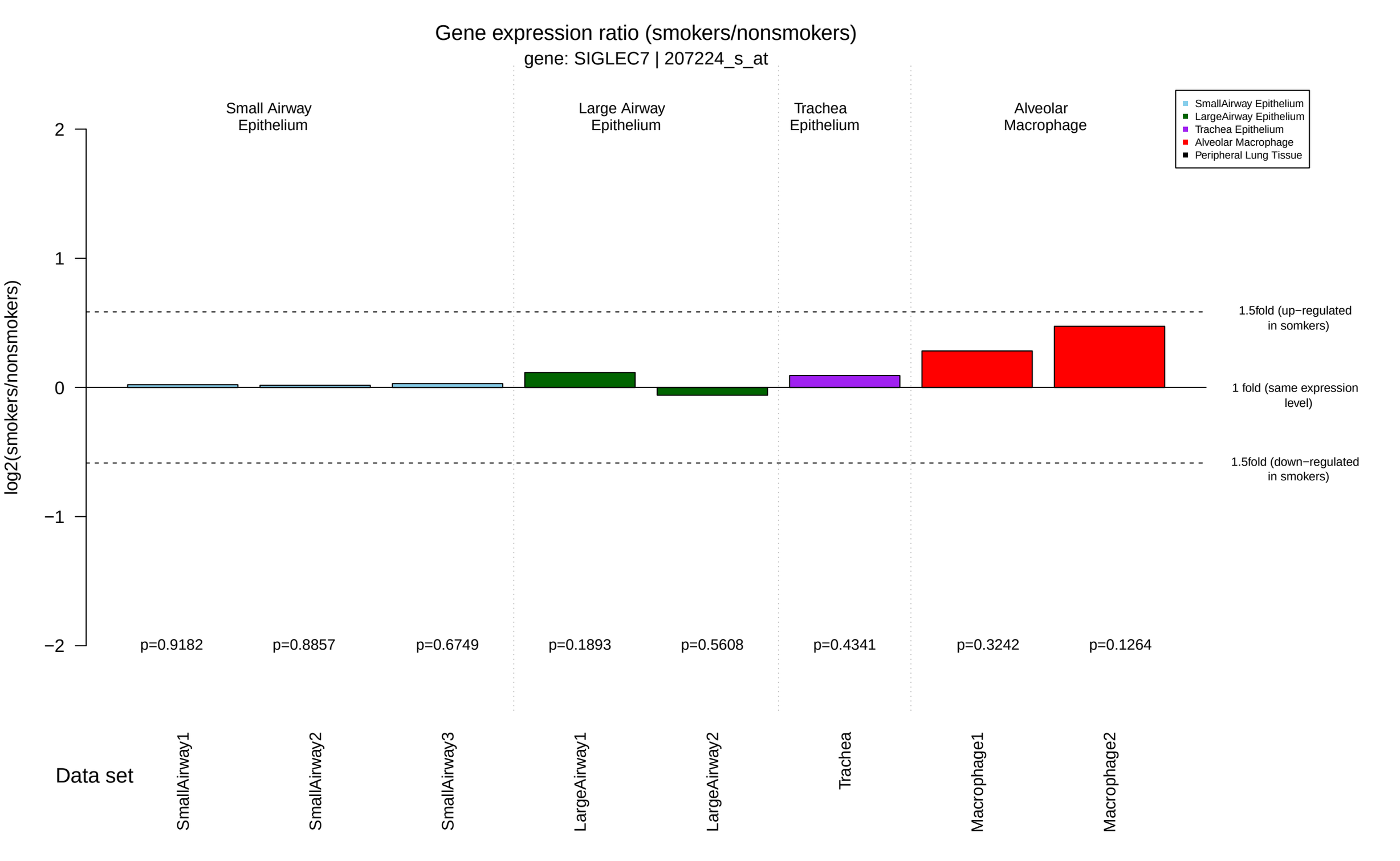
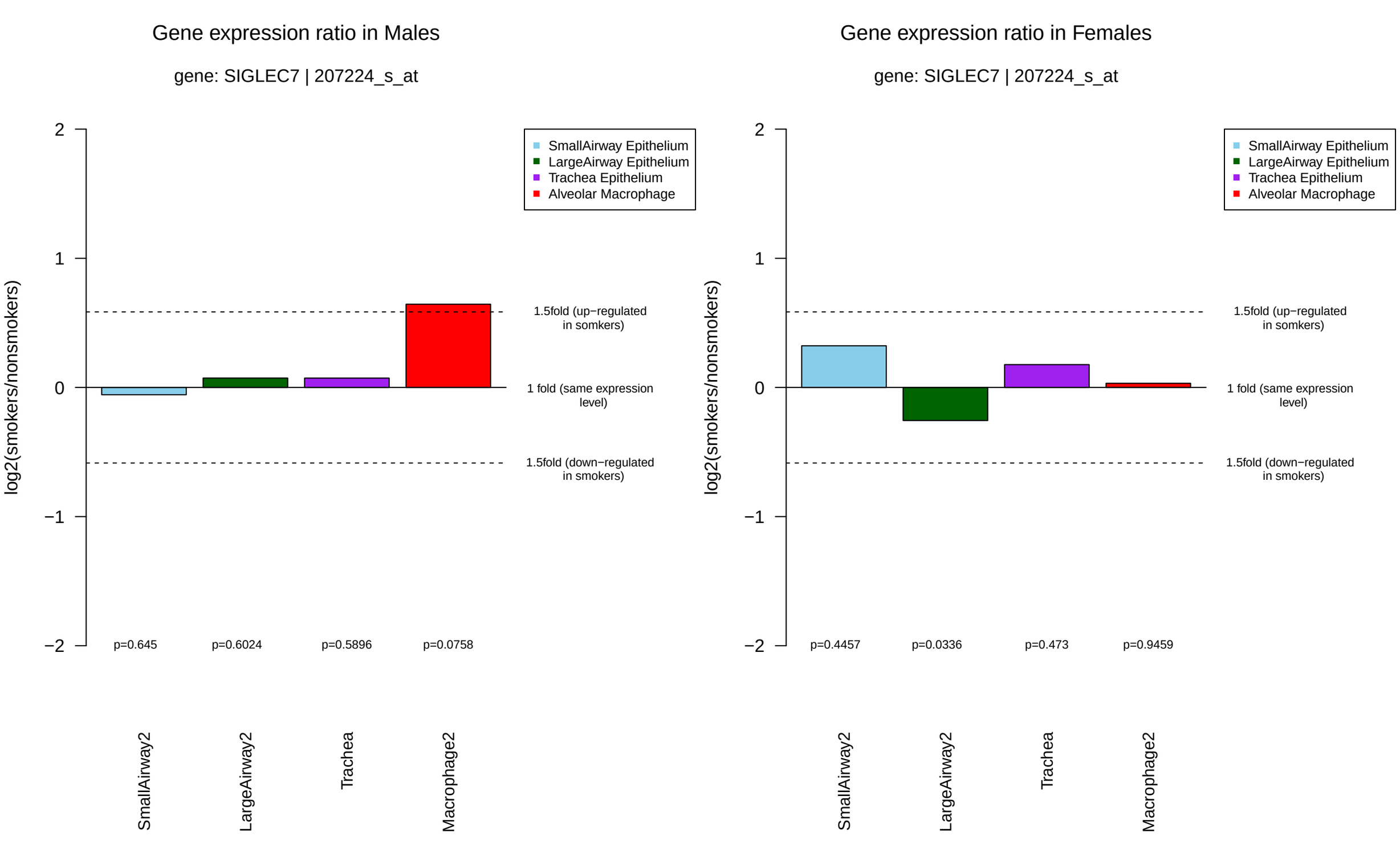
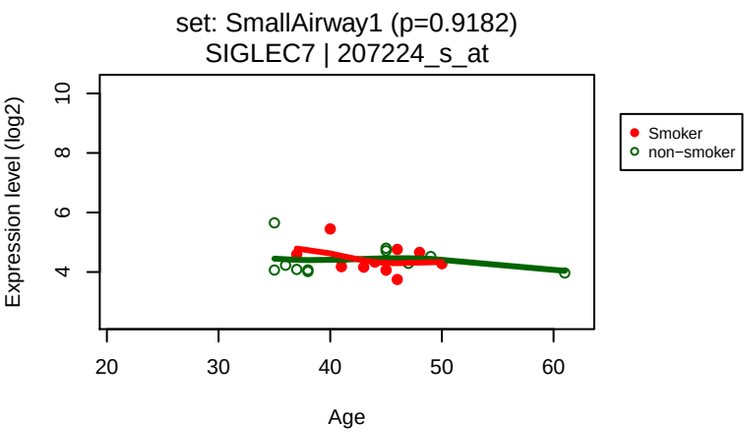
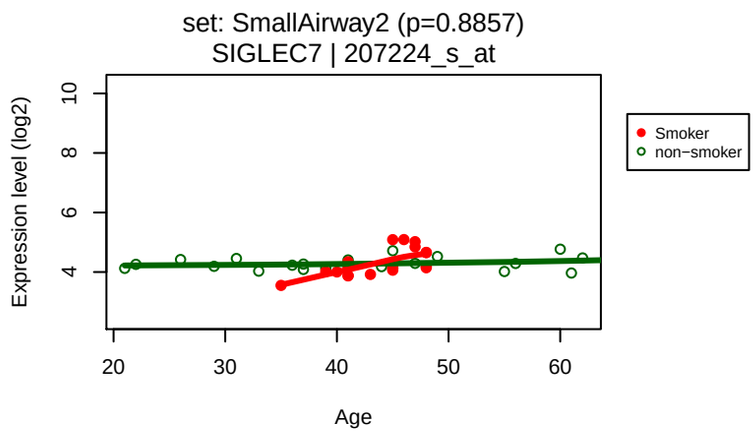
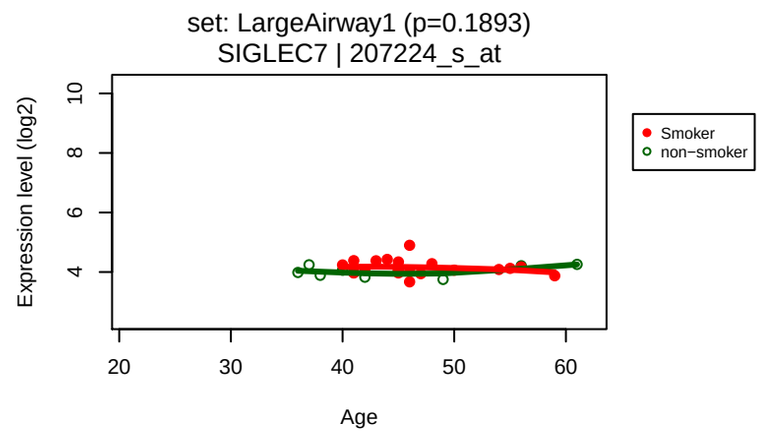

Gene expression of smokers and nonsmokers
gene: SIGLEC7 | 207224_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9182
p=0.8857
p=0.6749
p=0.1893
p=0.5608
p=0.4341
p=0.3242
p=0.1264
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

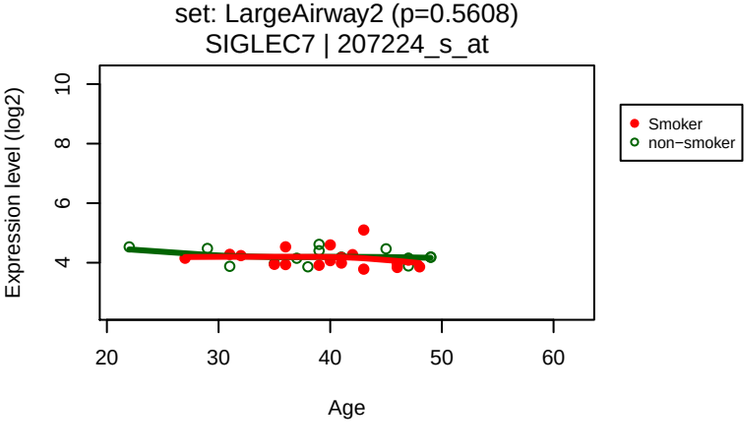

Gene expression ratio (smokers/nonsmokers)
gene: SIGLEC7 | 207224_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9182
p=0.8857
p=0.6749
p=0.1893
p=0.5608
p=0.4341
p=0.3242
p=0.1264
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

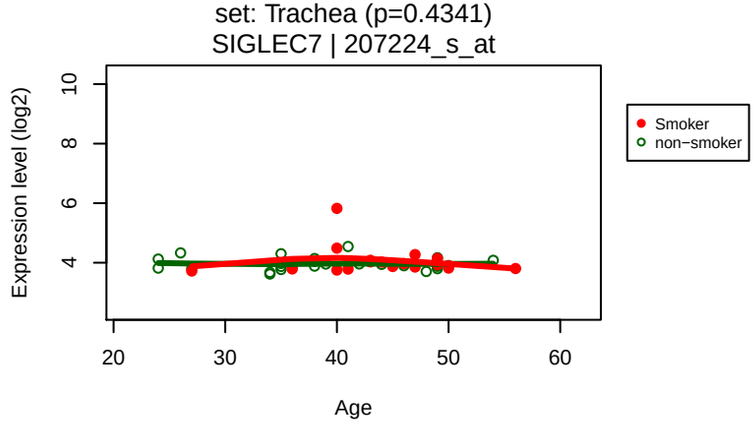

Gene expression ratio in Males
gene: SIGLEC7 | 207224_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.645
p=0.6024
p=0.5896
p=0.0758
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SIGLEC7 | 207224_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4457
p=0.0336
p=0.473
p=0.9459
SmallAirway2
LargeAirway2
Trachea
Macrophage2

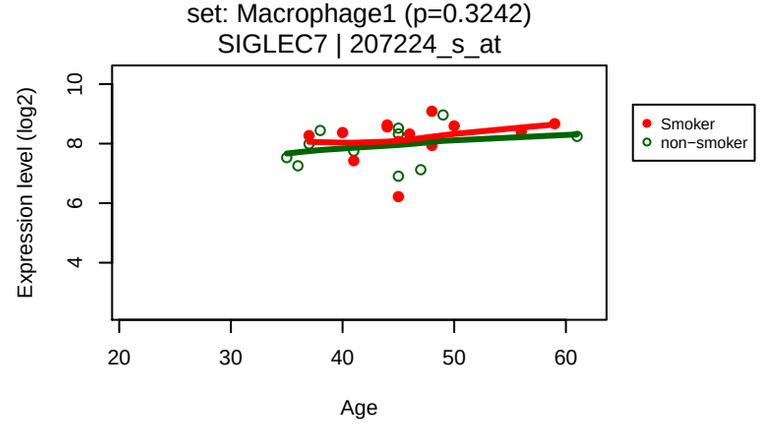

set: SmallAirway1 (p=0.9182)
SIGLEC7 | 207224_s_at
Expression level (log2)
Age
Smoker
non−smoker

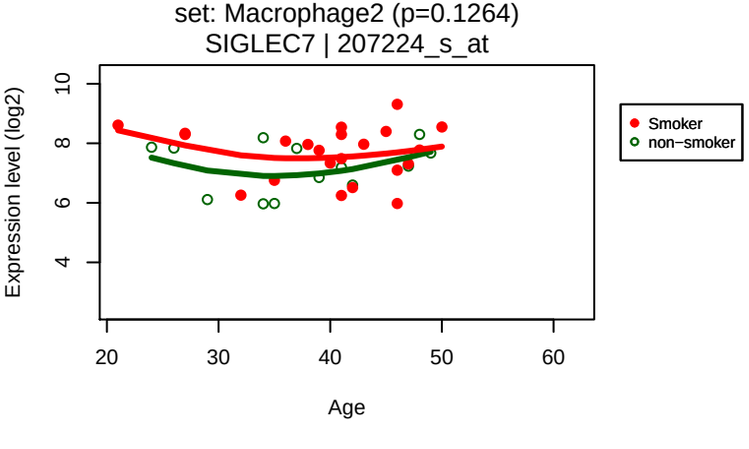

set: SmallAirway2 (p=0.8857)
SIGLEC7 | 207224_s_at
Expression level (log2)
Age
Smoker
non−smoker

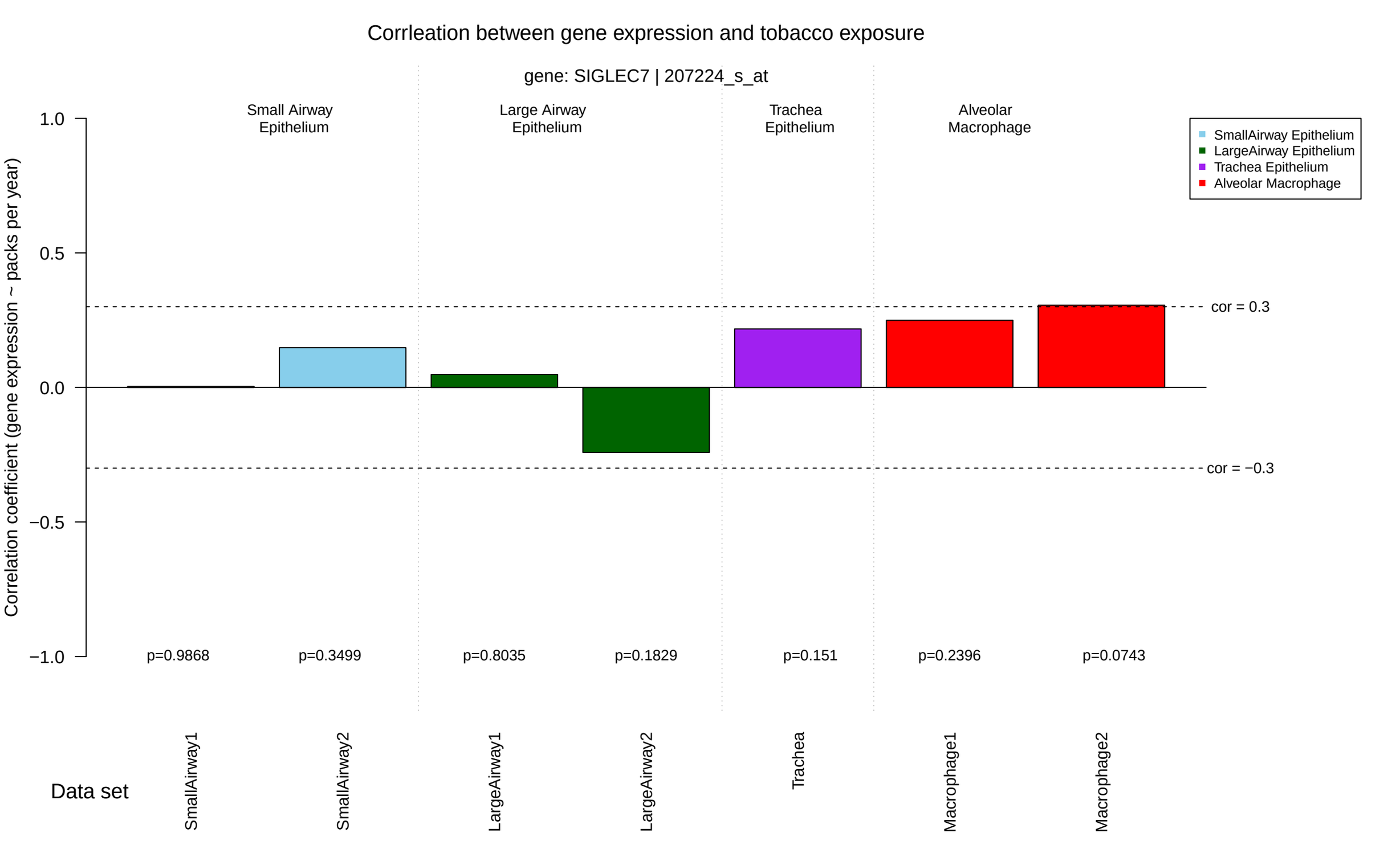

set: LargeAirway1 (p=0.1893)
SIGLEC7 | 207224_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5608)
SIGLEC7 | 207224_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4341)
SIGLEC7 | 207224_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3242)
SIGLEC7 | 207224_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1264)
SIGLEC7 | 207224_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SIGLEC7 | 207224_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9868
p=0.3499
p=0.8035
p=0.1829
p=0.151
p=0.2396
p=0.0743
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2